



Redshift Evolution of Electron Density in the Interstellar Medium at $z \sim 0-9$ Uncovered with JWST/NIRSpec Spectra and Line-spread Function Determinations

Yuki Isobe^{1,2} , Masami Ouchi^{1,3,4} , Kimihiko Nakajima³ , Yuichi Harikane¹ , Yoshiaki Ono¹ , Yi Xu^{1,5} ,
Yechi Zhang^{1,5} , and Hiroya Umeda^{1,2}

¹ Institute for Cosmic Ray Research, The University of Tokyo, 5-1-5 Kashiwanoha, Kashiwa, Chiba 277-8582, Japan; isobe@icrr.u-tokyo.ac.jp

² Department of Physics, Graduate School of Science, The University of Tokyo, 7-3-1 Hongo, Bunkyo, Tokyo 113-0033, Japan

³ National Astronomical Observatory of Japan, 2-21-1 Osawa, Mitaka, Tokyo 181-8588, Japan

⁴ Kavli Institute for the Physics and Mathematics of the Universe (WPI), University of Tokyo, Kashiwa, Chiba 277-8583, Japan

⁵ Department of Astronomy, Graduate School of Science, The University of Tokyo, 7-3-1 Hongo, Bunkyo, Tokyo 113-0033, Japan

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Abstract

We present electron densities n_e in the interstellar medium (ISM) of star-forming galaxies at $z = 4-9$ observed by the JWST/NIRSpec GLASS, Early Release Observations, and CEERS programs. We carefully evaluate the line-spread functions of the NIRSpec instrument as a function of wavelength with the calibration data of a planetary nebula taken on board, and obtain secure [O II] $\lambda\lambda 3726, 3729$ doublet fluxes for 14 galaxies at $z = 4.02-8.68$ falling on the star formation main sequence with the NIRSpec high- and medium-resolution spectra. We thus derive the electron densities of singly ionized oxygen nebulae with the standard n_e indicator of the [O II] doublet, and find that the electron densities of the $z = 4-9$ galaxies are $n_e \gtrsim 300 \text{ cm}^{-3}$, significantly higher than those of low- z galaxies at a given stellar mass, star formation rate (SFR), and specific SFR. Interestingly, the typical electron densities of the singly ionized nebulae increase from $z = 0$ to $z = 1-3$ and $z = 4-9$, which is approximated by the evolutionary relation of $n_e \propto (1+z)^p$ with $p \sim 1-2$. Although it is not obvious that the ISM property of n_e is influenced by global galaxy properties, these results may suggest that the nebula densities of high- z galaxies are generally high due to the compact morphologies of high- z galaxies evolving by $r_e \propto (1+z)^{-1}$ ($r_{\text{vir}} \propto (1+z)^{-1}$) for a given stellar (halo) mass whose inverse square corresponds to the $p \sim 2$ evolutionary relation. The $p \sim 1-2$ evolutionary relation can be explained by a combination of the compact morphology and the reduction of n_e due to the high electron temperature of high- z metal-poor nebulae.

Unified Astronomy Thesaurus concepts: Galaxy formation (595); Galaxy structure (622); Star formation (1569); Dwarf galaxies (416)

1. Introduction

Properties of the interstellar medium (ISM) have been spectroscopically investigated by many studies using photo-ionization models (e.g., Nagao et al. 2012; Nakajima & Ouchi 2014; Feltre et al. 2016; Sanders et al. 2016a; Steidel et al. 2016; Schaerer et al. 2018; Kewley et al. 2019; Harikane et al. 2020; Berg et al. 2021; Matsumoto et al. 2022; Umeda et al. 2022). In particular, the electron densities n_e of H II regions are one of the key quantities to describe the physical states of the ISM. The electron density is usually derived from density-sensitive emission-line ratios such as [O II] $\lambda 3726$ /[O II] $\lambda 3729$, [S II] $\lambda 6716$ /[S II] $\lambda 6731$, and [C III] $\lambda 1907$ /[C III] $\lambda 1909$ (e.g., Kewley et al. 2019). Large spectroscopic surveys have found that star-forming galaxies with $M_* \sim 10^{10} M_\odot$ have n_e values of low-ionization regions (O II and S II) increasing from 30 to $\sim 200 \text{ cm}^{-3}$ with the redshift in the range of $z \sim 0-3$ (e.g., Steidel et al. 2014; Shimakawa et al. 2015; Sanders et al. 2016a; Kashino et al. 2017; Davies et al. 2021). At $z \gtrsim 4$, n_e values had not been investigated due to the lack of high-sensitivity near-infrared spectrographs before the arrival of the James Webb Space Telescope (JWST), except for a few suggestions based on doubly ionized regions such as $n_e(\text{C III}) \sim 4-13 \times 10^4 \text{ cm}^{-3}$ at $z = 10.957$ (Jiang et al. 2021)

and $n_e(\text{O III}) \lesssim 500 \text{ cm}^{-3}$ at $z = 7.133$ (Killi et al. 2023). Recently, using the flux ratio of [O III] $\lambda 5007$ from the JWST/NIRSpec Early Release Observations (ERO; Pontoppidan et al. 2022) and [O III] $88 \mu\text{m}$ from Atacama Large Millimeter/submillimeter Array observations, Fujimoto et al. (2022) have reported that a $z = 8.496$ dwarf galaxy, s04590, has an $n_e(\text{O III})$ value of $220^{+170}_{-100} \text{ cm}^{-3}$.

Although the $n_e(\text{O III})$ or C III values are likely to be higher than those of $z \sim 0$ star-forming galaxies (e.g., $n_e(\text{O II}) \sim 30 \text{ cm}^{-3}$; Davies et al. 2021), we need to discuss the redshift evolution of n_e at similar-ionization regions (i.e., O II and S II, rather than O III or C III). We should also check if n_e values change with other galaxy properties such as M_* or the star formation rate (SFR). In this paper, we aim to measure the $n_e(\text{O II})$ values of high- z ($z > 4$) star-forming galaxies identified by NIRSpec public surveys. Comparing the n_e values with those of lower- z star-forming galaxies, we also investigate the redshift evolution of n_e and the potential dependences on M_* and SFR.

This paper is organized as follows. Section 2 explains our data and sample. Our analysis is described in Section 3. We report our results and discuss them in Section 4. Our findings are summarized in Section 5. We assume a standard ΛCDM cosmology with parameters of $(\Omega_m, \Omega_\Lambda, H_0) = (0.3, 0.7, 70 \text{ km s}^{-1} \text{ Mpc}^{-1})$.

2. Data and Sample

We use the JWST/NIRSpec data of the ERO (Pontoppidan et al. 2022) taken in the SMACS 0723 lensing cluster field

(hereafter the ERO data), GLASS (Treu et al. 2022; hereafter the GLASS data), and CEERS (Finkelstein et al. 2023; hereafter the CEERS data). The ERO data were taken with the medium-resolution ($R \sim 1000$) filter-grating pairs F170LP–G235M and F290LP–G395M covering the wavelength ranges of 1.7–3.1 and 2.9–5.1 μm , respectively. The total exposure time of the ERO data is 4.86 hr for each filter-grating pair. The GLASS data were taken with the high-resolution ($R \sim 2700$) filter-grating pairs F100LP–G140H, F170LP–G235H, and F290LP–G395H covering the wavelength ranges of 1.0–1.6, 1.7–3.1, and 2.9–5.1 μm , respectively. The total exposure time of the GLASS data is 4.9 hr for each filter-grating pair. The CEERS data were taken with the medium-resolution filter-grating pairs F100LP–G140M, F170LP–G235M, and F290LP–G395M covering the wavelength ranges of 1.0–1.6, 1.7–3.1, and 2.9–5.1 μm , respectively. The total exposure time of the CEERS data is 0.86 hr for each filter-grating pair.

We use spectroscopic data reduced by Nakajima et al. (2023). Nakajima et al. (2023) took the raw data from the Mikulski Archive for Space Telescopes (MAST) and conducted level 2 and 3 calibrations, using the JWST Science Calibration Pipeline with the reference file `jwst_1028.pmap`, whose flux calibration is based on in-flight flat data. In addition to the readout noise and Poisson noise outputted by the JWST/NIRSpec pipeline, Nakajima et al. (2023) added the standard deviation of the residual background to evaluate the uncertainty of background subtraction. Checking the data, we identify 5, 14, and 55 galaxies at $z > 4$ in the ERO, GLASS, and CEERS data, respectively, whose redshifts were obtained by simultaneous fitting of H β and [O III] $\lambda\lambda 4959, 5007$. We omit galaxies with signal-to-noise ratios (S/N) of the [O II] doublet less than 10 so that we can obtain galaxies with well-determined n_e . After the selection, we obtain 1, 4, and 11 galaxies from the ERO, GLASS, and CEERS data, respectively. Hereafter we refer to the 16 (=1 + 4 + 11) galaxies as the JWST galaxies.

3. Analysis

In this paper, we measure [O II] $\lambda 3726$ /[O II] $\lambda 3729$ ratios to obtain $n_e(\text{O II})$. The separation of the observed wavelengths of the [O II] $\lambda\lambda 3726, 3729$ doublet is even shorter than double the expected FWHM of the $R \sim 2700$ grating, which means that the observed [O II] doublet is not fully deblended even with the $R \sim 2700$ grating. To deblend the [O II] doublet, we should determine emission-line profiles of observed galaxies using the line-spread functions (LSFs) of NIRSpec and the intrinsic velocity dispersions (σ_{int}) of the observed galaxies. Section 3.1 describes how we derive the LSFs, and Section 3.2 explains how we measure σ_{int} .

3.1. LSF

We obtain the LSFs from a planetary nebula (PN), IRAS-05248-7007. NIRSpec spectra of the PN were taken during the commissioning process (proposal ID: 1125). The PN was observed with a slit whose width is 0".2, which is the same as those of the MSA shutters. We use the level 3 spectra available at MAST. We choose emission lines with high S/N. We also avoid using emission lines blended with other nearby features. After the selection, we obtain 4, 11, and 5 emission lines for the spectra obtained with the filters F100LP, F170LP, and F290LP, respectively. The selected emission lines cover wide wavelength ranges of 1.1–1.7, 1.8–2.8, and 3.1–4.5 μm for the filters

F100LP, F170LP, and F290LP, respectively. To obtain the LSF at an arbitrary wavelength, we interpolate the LSF profiles.

Figure 1 summarizes the LSFs at different wavelengths for each filter-grating pair. The obtained LSFs of F100LP–G140H, F170LP–G235H, F290LP–G395H, F100LP–G140M, F170LP–G235M, and F290LP–G395M have median resolution values of $R \simeq 3180, 2800, 3390, 1160, 1120$, and 1080 obtained in wavelength ranges of 1.1–1.7, 1.8–2.8, 3.1–4.5, 1.1–1.7, 1.8–2.8, and 3.1–4.5 μm , respectively. These resolution values are $\sim 10\%$ – 20% larger than the nominal resolutions of $R \sim 2700$ for the high-resolution gratings and $R \sim 1000$ for the medium-resolution gratings. We also confirm that the LSFs of each filter-grating pair become narrower (i.e., represent better resolution) with longer wavelengths, which is also expected by the NIRSpec instrumental team.⁶

The PN was also observed by VLT/X-shooter with the spectral resolution of $R \sim 6500$ (Euclid Collaboration et al. 2023; the spectral properties are available in the VizieR atlas⁷). The observed line widths of the PN are nearly the same as those expected from the spectral resolution of 6500, which suggests that the PN's bulk motions are not resolved with $R \sim 6500$. This also indicates that NIRSpec cannot resolve the PN's bulk motions with the NIRSpec spectral resolution of ~ 1000 or ~ 2700 .

Therefore, the PN may have unresolved bulk motions. Given that the intrinsic velocity dispersions (σ_{int}) of PNs are typically $\sim 30 \text{ km s}^{-1}$ (Jacob et al. 2013), we may not be able to measure the σ_{int} values of galaxies with $\sigma_{\text{int}} \lesssim 30 \text{ km s}^{-1}$. To put it another way, the σ_{int} value of 30 km s^{-1} does not change the LSF profiles significantly, i.e., σ_{int} accounts for only $\sim 1\%$ and $\sim 8\%$ of the observed velocity dispersion of the PN with the $R \sim 1000$ and ~ 2700 gratings, respectively.

3.2. Velocity Dispersion Measurement

To obtain the σ_{int} of the observed galaxies, we fit a Gaussian function convolved with the LSF derived in Section 3.1. To obtain σ_{int} with smaller uncertainties, we measure the σ_{int} of [O III] $\lambda 5007$, which is the strongest in the observed emission lines. We conduct the fittings to [O III] $\lambda\lambda 4959, 5007$ simultaneously so that the line profiles of the two emission lines are reproduced self-consistently. We fix the flux ratio [O III] $\lambda 5007$ /[O III] $\lambda 4959$ to 2.98, which is accurately determined by the Einstein A coefficient (Storey & Zeippen 2000).

We note that our fitting provides $\sigma_{\text{int}} \sim 0 \text{ km s}^{-1}$ for 2 of the 16 JWST galaxies (CEERS_01173 and CEERS_01539). We adopt $\sigma_{\text{int}} = 0 \text{ km s}^{-1}$ for the two JWST galaxies to determine the [O II] doublet profiles, while the true σ_{int} values of the two JWST galaxies are likely to be larger than 0 km s^{-1} and less than that of the PN used for the LSFs (i.e., $0 < \sigma_{\text{int, true}} \lesssim 30 \text{ km s}^{-1}$; Section 3.1).

3.3. Electron Density Measurement

Using the LSF (Section 3.1), we measure the fluxes of the [O II] doublet. We fit three Gaussian functions convolved by the LSF to [O II] $\lambda\lambda 3726, 3729$, and [O III] $\lambda 5007$ simultaneously with the fixed z . Figure 2 shows our fitting results. We

⁶ <https://jwst-docs.stsci.edu/jwst-near-infrared-spectrograph/nirspec-instrumentation/nirspec-dispersers-and-filters>

⁷ http://vizier.cfa.harvard.edu/viz-bin/VizieR-3?-source=J/A%2bA/674/A172/pn-0-11&-out.max=50&-out.form=HTML%20Table&-out.add=_r&-out.add=_RAJ,_DEJ&-sort=_r&-oc.form=sexa

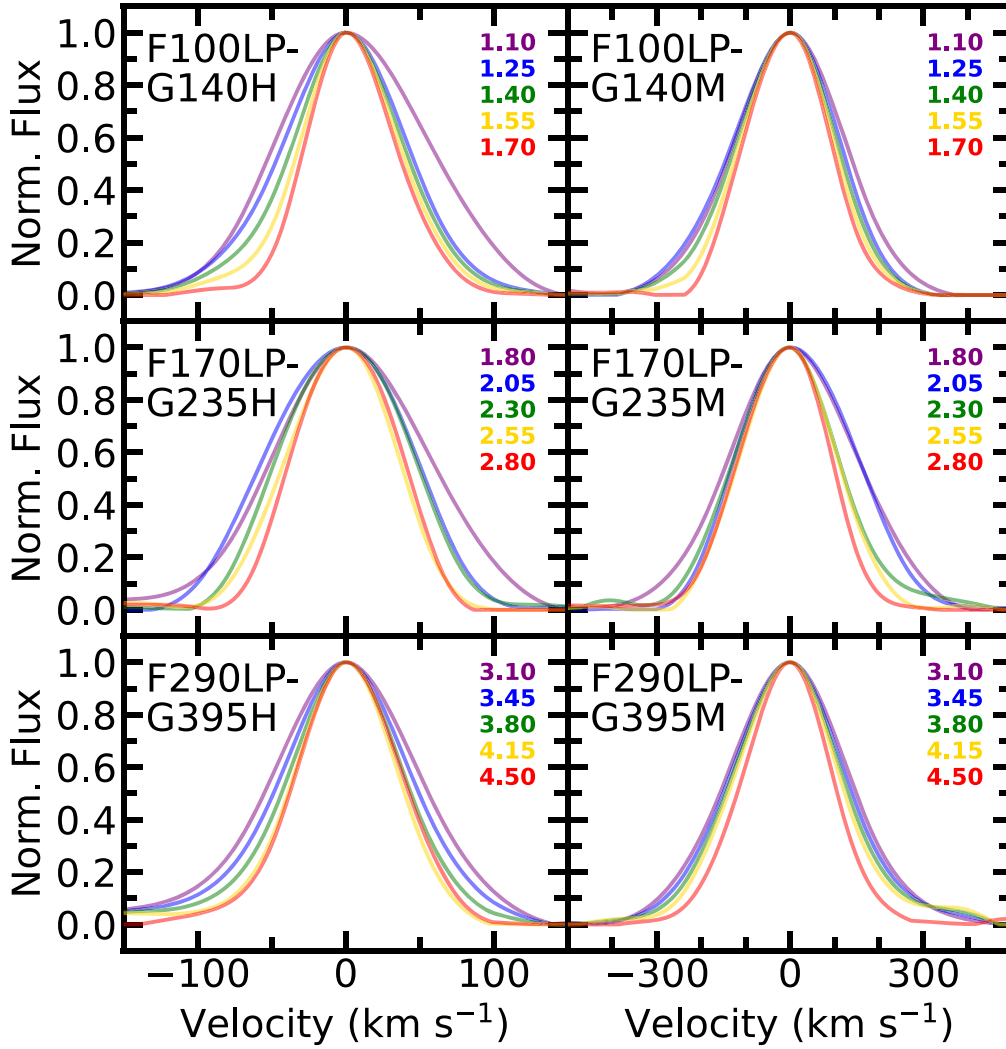


Figure 1. LSF as a function of velocity. In each panel, the purple, blue, green, yellow, and red curves represent LSFs obtained at wavelengths whose values correspond to the numbers with the same color listed in the top right corner (units: μm).

use **PyNeb** (Luridiana et al. 2015; v1.1.15) to obtain n_e from the [O II] doublet. This method provides n_e values consistent with those derived from the updated relation between [O II] $\lambda 3729$ /[O II] $\lambda 3726$ and the n_e reported by Sanders et al. (2016a). In ERO_06355, GLASS_150029, GLASS_160133, and CEERS_00397, electron temperatures of 11,300, 17,600, 14,800, and 15,200 K, respectively, have been measured from the [O III] $\lambda 4363$ line (Isobe et al. 2023; Nakajima et al. 2023). For these four galaxies, we use these temperature values when deriving the n_e values. For the other JWST galaxies, we assume electron temperatures of 15,000 K, which is typical for star-forming dwarf galaxies (e.g., Izotov et al. 2012; Kojima et al. 2020; Isobe et al. 2022; Nakajima et al. 2022). We perform Monte Carlo simulations based on spectral errors to obtain uncertainties of n_e . We note that 2 of the 16 JWST galaxies (GLASS_10021 and CEERS_01449) show unphysical [O II] $\lambda 3726$ /[O II] $\lambda 3729$ ratios, even when including the spectral errors into the n_e measurements. Omitting the two galaxies, we obtain a final sample consisting of 14 ($=16 - 2$) galaxies. Table 1 lists the derived n_e with the uncertainties.

Since the n_e values based on the $R \sim 2700$ observations are slightly lower than those with $R \sim 1000$, we perform a recovery test for the $R \sim 1000$ data by making mock spectra around the

[O II] $\lambda \lambda 3726, 3729$ and [O III] $\lambda 5007$ lines with a typical σ_{int} of 60 km s^{-1} convolved by LSFs and noises that match the typical S/N of these line fluxes. Conducting line fitting to the mock spectra in the same way as for the scientific data, we confirm that our fitting method can reproduce electron density values within at most 3.5% accuracy even for the $R \sim 1000$ spectra.

We check that different assumptions of electron temperatures from $T_e = 10,000$ to $30,000$ K can change n_e values by at most $\sim 20\%$ from those calculated by the original assumption of $T_e = 15,000$ K. These systematics of n_e are much smaller than those derived from the spectral errors.

3.4. M_* and SFR

In this paper, we characterize the JWST galaxies with their M_* and SFR values. Seven of the 14 JWST galaxies have JWST/NIRCam photometry at the time of writing. Regarding these seven galaxies, we use the M_* and SFR values derived by Nakajima et al. (2023). Nakajima et al. (2023) derived M_* by conducting SED fitting with **prospector** (Johnson et al. 2021) assuming the Chabrier (2003) initial mass function (IMF), a flexible star formation history, and the Calzetti et al. (2000) dust extinction law, in a similar manner to Harikane et al. (2023).

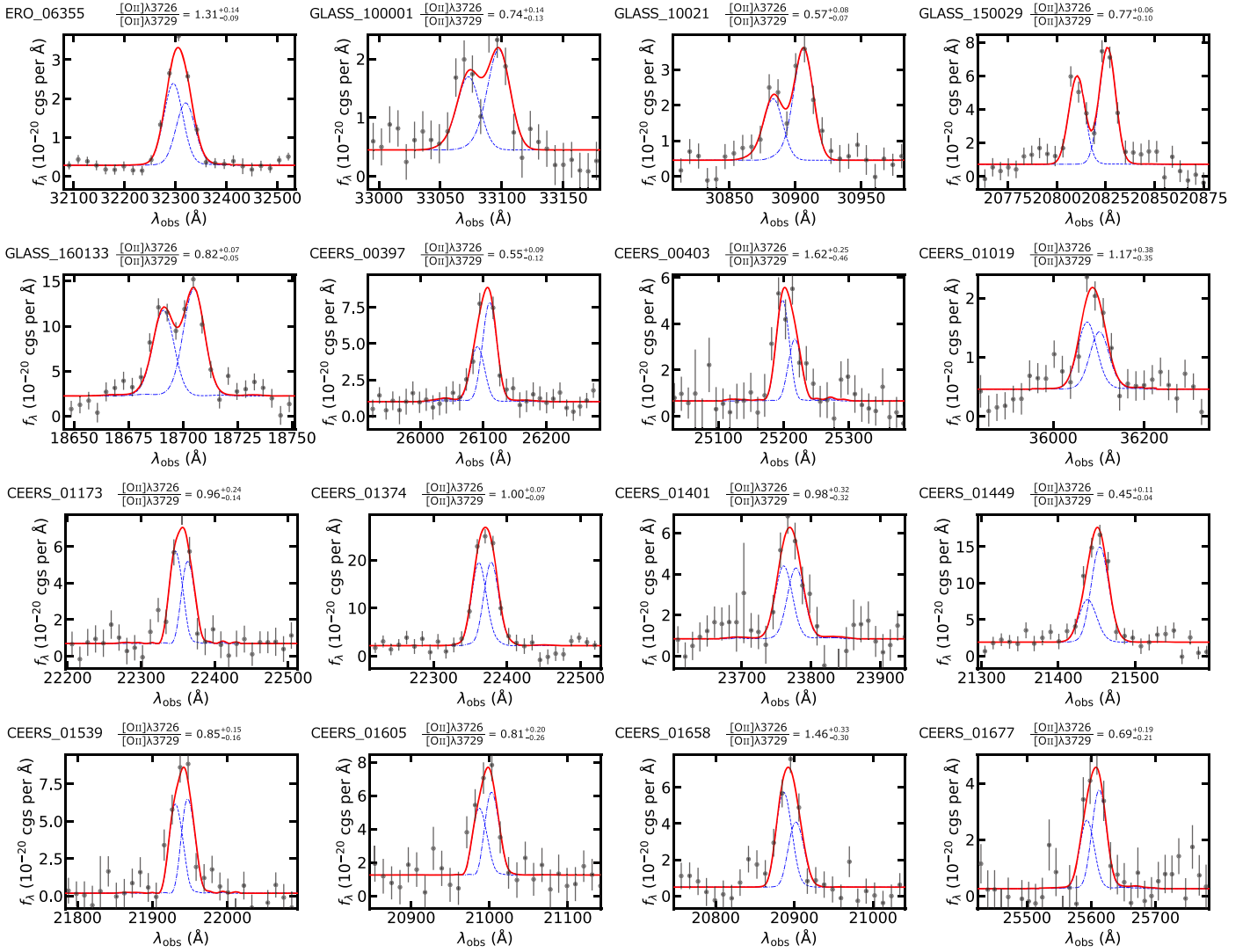


Figure 2. Our fitting results of the LSF-convolved Gaussian functions to the [O II] doublet. The gray dots denote data points with spectral errors. The red solid curve represents the total profile of the [O II] doublet. The blue dashed curve and the blue dashed-dotted curve show the profiles of [O II] $\lambda 3726$ and [O II] $\lambda 3729$, respectively.

NIRCam photometries of the JWST galaxies are measured by Source Extractor (Bertin & Arnouts 1996) as in Harikane et al. (2023). The photometries are corrected for gravitational lensing magnifications as modeled by *glafic* (Oguri 2010). We then correct the NIRSpectra for slit losses by scaling the spectra to match the NIRCam photometries, where the typical correction factor is about 1.3. The SFR values are calculated from slit loss-corrected $H\beta$ luminosities with the NIRCam rest-frame optical photometry under the assumptions of the Kennicutt (1998) relation and Chabrier (2003) IMF. Figure 3 shows the M_* -SFR relations of the JWST galaxies with the NIRCam photometry. The median M_* and SFR values of the JWST galaxies (large red circle) show that the JWST galaxies generally lie on the star formation main sequence (SFMS) at $z=5-6$ (Santini et al. 2017).

We estimate the M_* of the remaining 7 ($=14 - 7$) JWST galaxies from broadband photometry of HST/Advanced Camera for Surveys F606W or F814W or HST/WFC3 F125W taken from the CANDELS Extended Groth Strip catalog (Stefanon et al. 2017). We use FLUX_AUTO values based on Source Extractor (Bertin & Arnouts 1996) as the total

fluxes. Assuming that these seven JWST galaxies lie on the SFMS like the seven JWST galaxies with NIRCam photometry, we adopt empirical relations between M_* and the absolute magnitude of rest frame $1500 \text{ \AA}$ (M_{UV}) derived at $z=4, 5, 6, 7$, and 8 (Song et al. 2016; based on the Salpeter 1955 IMF). We choose the M_* - M_{UV} relation and HST photometry consistent with the redshift of each JWST galaxy. Note that three of the seven JWST galaxies do not have $>5\sigma$ detection of HST photometry. For these three JWST galaxies, we calculate upper limits of M_* based on 5σ limiting magnitudes of the HST photometry. For consistency with the M_* values derived by Nakajima et al. (2023), we rescale the M_* values from the Salpeter (1955) IMF to the Chabrier (2003) IMF by multiplying them by 0.61 (Madau & Dickinson 2014). We check that the M_* values based on the HST photometry are consistent with those of Nakajima et al. (2023) within a 1σ level. We propagate errors of the HST photometry and the M_* - M_{UV} relations to the final stellar mass errors. The scatters of the M_* - M_{UV} relations are ~ 0.15 dex at $z \sim 4$ and ~ 0.3 dex at $z \sim 8$, which are smaller than those of our derived stellar

Table 1
Electron Density and Other Properties

Name	R.A.	Decl.	z	$n_e(\text{O II})$	$\frac{[\text{O II}]\lambda 3726}{[\text{O II}]\lambda 3729}$	σ_{int}	$\log(M_*)$	$\log(\text{SFR})$
(1)	(deg)	(deg)	(4)	(cm $^{-3}$)	(6)	(km s $^{-1}$)	($M_\odot$)	($M_\odot \text{ yr}^{-1}$)
ERO_06355	110.8267392	-73.4537444	7.6651	1433^{+752}_{-335}	$1.31^{+0.14}_{-0.09}$	113	$8.77^{+0.08a}_{-0.01}$	$1.41^{+0.01a}_{-0.01}$
GLASS_100001	3.6038450	-30.3822350	7.8737	118^{+217}_{-118}	$0.74^{+0.14}_{-0.13}$	69	$9.15^{+0.55a}_{-0.00}$	$0.95^{+0.10a}_{-0.13}$
GLASS_150029	3.5771664	-30.4225760	4.5837	174^{+137}_{-174}	$0.77^{+0.06}_{-0.10}$	38	$9.12^{+0.03a}_{-0.33}$	$1.04^{+0.02a}_{-0.02}$
GLASS_160133	3.5802754	-30.4244040	4.0150	252^{+161}_{-103}	$0.82^{+0.07}_{-0.05}$	59	$8.11^{+0.41a}_{-0.22}$	$1.16^{+0.01a}_{-0.01}$
CEERS_00397	214.8361970	52.8826930	6.0005	0^{+1}_{-0}	$0.55^{+0.09}_{-0.12}$	81	$8.45^{+0.53a}_{-0.06}$	$1.76^{+0.02a}_{-0.03}$
CEERS_00403	214.8289680	52.8757010	5.7609	3105^{+3232}_{-2151}	$1.62^{+0.25}_{-0.46}$	56	$9.35^{+0.27a}_{-0.09}$	$1.20^{+0.06a}_{-0.07}$
CEERS_01019	215.0353914	52.8906618	8.6791	1118^{+3332}_{-969}	$1.17^{+0.38}_{-0.35}$	123	$10.12^{+0.12a}_{-0.11}$	$1.97^{+0.04a}_{-0.05}$
CEERS_01173	215.1542076	52.9558470	4.9957	574^{+942}_{-331}	$0.96^{+0.24}_{-0.14}$	... ^c	$9.38^{+1.12b}_{-1.12}$	$0.56^{+0.13b}_{-0.13}$
CEERS_01374	214.9439110	52.8500419	4.9999	656^{+254}_{-219}	$1.00^{+0.07}_{-0.09}$	67	$9.44^{+1.12b}_{-1.12}$	$1.36^{+0.03b}_{-0.03}$
CEERS_01401	215.2458005	53.0652953	5.3749	613^{+1934}_{-613}	$0.98^{+0.32}_{-0.32}$	112	$<8.92^b$	$0.86^{+0.09b}_{-0.09}$
CEERS_01539	214.9800779	52.9426590	4.8841	330^{+444}_{-327}	$0.85^{+0.15}_{-0.16}$	... ^c	$<8.71^b$	$0.80^{+0.05b}_{-0.05}$
CEERS_01605	215.0754073	52.9975786	4.6309	250^{+537}_{-250}	$0.81^{+0.20}_{-0.26}$	16	$9.11^{+1.10b}_{-1.10}$	$0.53^{+0.06b}_{-0.06}$
CEERS_01658	214.9852372	52.9242590	4.6038	2199^{+5950}_{-1078}	$1.46^{+0.33}_{-0.30}$	64	$<8.56^b$	$0.67^{+0.06b}_{-0.06}$
CEERS_01677	215.1887384	53.0643782	5.8670	39^{+470}_{-39}	$0.69^{+0.19}_{-0.21}$	83	$9.07^{+0.84b}_{-0.84}$	$0.69^{+0.08b}_{-0.08}$

Notes. (1) Name. (2) R.A. in J2000. (3) Decl. in J2000. (4) Redshift. (5) Electron density derived from the [O II] doublet (Section 3.3). (6) [O II] λ 3726/[O II] λ 3729 ratio. (7) Velocity dispersion (Section 3.2). (8) Stellar mass (Section 3.4). (9) SFR (Section 3.4).

^a The M_* and SFR values are estimated from spectral energy distribution (SED) fitting to JWST/NIRCam photometry with prospector (Johnson et al. 2021) and H β fluxes corrected for slit losses, respectively (Nakajima et al. 2023; Section 3.4).

^b The M_* and SFR values are estimated from Hubble Space Telescope (HST) rest-frame UV photometry and H β fluxes not corrected for slit losses, respectively, because no NIRCam photometry is available for these objects to date (Section 3.4).

^c The σ_{int} values are too small to measure (Sections 3.1 and 3.2).

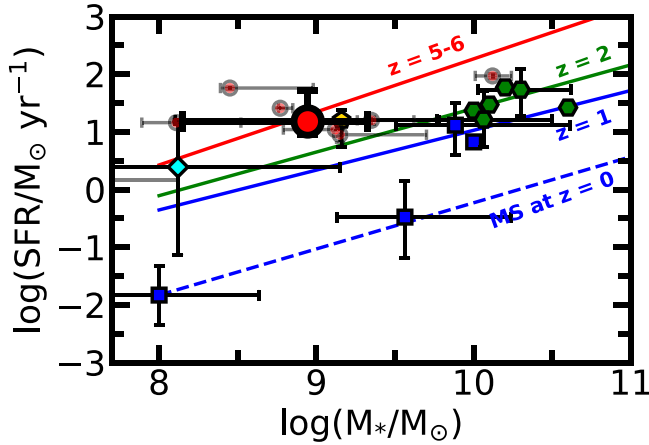


Figure 3. M_* -SFR relation of the JWST galaxies with the NIRCam photometry at $4 < z < 9$ represented by the red circles. We also plot the median n_e values of SFMS galaxies with $M_* \sim 10^8$ – $10^{10} M_\odot$ at $z \sim 0$ –1 (blue squares, [O II]- or [S II]-based; Berg et al. 2012; Swinbank et al. 2019; Davies et al. 2021) and with $M_* \sim 10^{10} M_\odot$ at $z \sim 1$ –3 (green hexagons, [O II]- or [S II]-based; Steidel et al. 2014; Sanders et al. 2016a; Kaasinen et al. 2017; Kashino et al. 2017; Davies et al. 2021). The red solid, green solid, blue solid, and blue dashed lines illustrate SFMS galaxies at $z = 5$ –6 (Santini et al. 2017), $z = 2$ (Speagle et al. 2014), $z = 1$ (Speagle et al. 2014), and $z = 0$ (Chang et al. 2015), respectively. We also show the median n_e values of high specific SFR (sSFR) galaxies with $M_* \sim 10^8 M_\odot$ at $z \sim 0$ (cyan diamond, [S II]-based; Berg et al. 2022) and with $M_* \sim 10^9 M_\odot$ at $z \sim 2$ –3 (yellow pentagon, [O II]-based; Christensen et al. 2012a, 2012b; Sanders et al. 2016b; Gburek et al. 2019).

mass errors. This indicates that the stellar mass errors are dominated by the HST photometry.

We also calculate the SFRs of the seven JWST galaxies from H β luminosities under the assumptions of the Kennicutt (1998) relation and Chabrier (2003) IMF in the same manner adopted

in Nakajima et al. (2023). Although we do not correct the H β luminosities of these seven galaxies without NIRCam photometry for slit losses, the small typical slit-loss correction factor of 1.3 suggests that the slit-loss correction does not change the SFR values much. We summarize the M_* and SFR values in Table 1.

4. Result and Discussion

Figure 4 presents the n_e - M_* (left), n_e -SFR (center), and n_e -sSFR (right) relations. The small red circles and the small red double circles in Figure 4 represent the properties of each JWST galaxy at $z \sim 4$ –6 and $z \sim 7$ –9, respectively. The large red circles and the large red double circles denote the median and 16th–84th percentiles of the properties of the JWST galaxies at $z \sim 4$ –6 and $z \sim 7$ –9, respectively. We find that the median n_e values of the JWST galaxies at $z \sim 4$ –6 and $z \sim 7$ –9 are ~ 300 and $\sim 1000 \text{ cm}^{-3}$, respectively.

For comparison, we compile the median n_e values of SFMS galaxies at $z \sim 0$ –1 (blue squares; Berg et al. 2012; Swinbank et al. 2019; Davies et al. 2021) and $z \sim 1$ –3 (green hexagons, [O II]- or [S II]-based; Steidel et al. 2014; Sanders et al. 2016a; Kaasinen et al. 2017; Kashino et al. 2017; Davies et al. 2021), whose M_* -SFR relations are shown in Figure 3. We note that we derive the median n_e value of Berg et al. (2012) from the flux ratios [S II] λ 6716/[S II] λ 6731 and T_e of their subsample of 31 galaxies. In addition, the median SFR values of Davies et al. (2021) are read from their figures.

It should be noted that n_e values are reported to be very dependent not on M_* (Kashino & Inoue 2019) but on SFR and sSFR (e.g., Shimakawa et al. 2015; Jiang et al. 2019; Kashino & Inoue 2019; Davies et al. 2021) at a given redshift. To examine how much the n_e values change with SFR and sSFR, we plot the 16th–84th percentile ranges of the n_e values of the

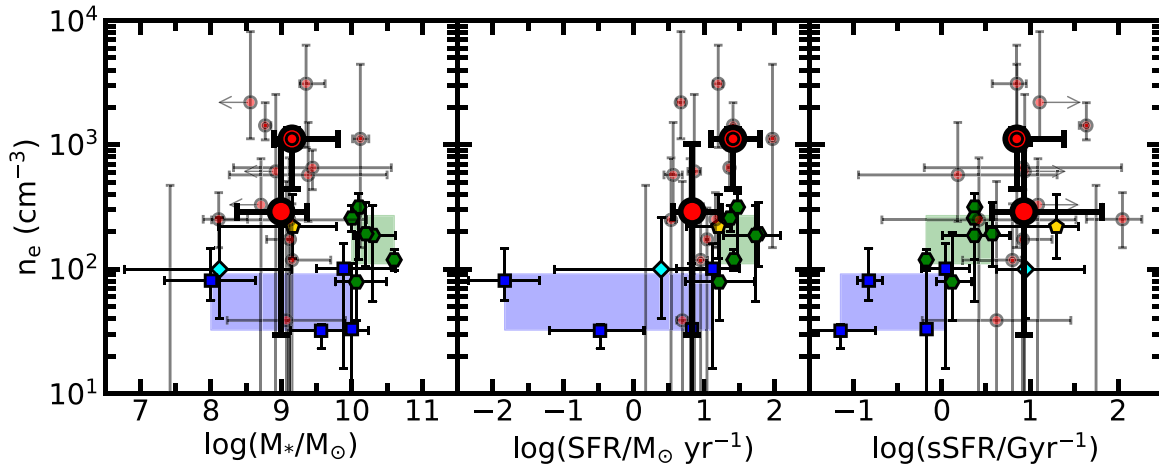


Figure 4. Electron density n_e as a function of M_* (left), SFR (center), and sSFR (right). The large red circles and the large red double circles denote the median and 16th–84th percentiles of the properties of the JWST galaxies at $z \sim 4$ –6 and $z \sim 7$ –9, respectively. The small red circles and the small red double circles represent the properties of each JWST galaxy at $z \sim 4$ –6 and $z \sim 7$ –9, respectively. The other symbols are the same as those in Figure 3. The blue and green shaded regions correspond to the 16th–84th percentile ranges of the n_e values of the galaxies at $z \sim 0$ –1 (Berg et al. 2012; Swinbank et al. 2019; Davies et al. 2021) and $z \sim 1$ –3 (Steidel et al. 2014; 2016a; Kaasinen et al. 2017; Kashino et al. 2017; Davies et al. 2021), respectively. On average, the JWST galaxies have higher n_e than lower- z galaxies at a given M_* , SFR, and sSFR.

SFMS galaxies at $z \sim 0$ –1 with a blue shaded region and at $z \sim 1$ –3 with a green shaded region in Figure 4. The blue shaded region indicates that the n_e dependencies on M_* , SFR, and sSFR in the ranges of $\log(M_*/M_\odot) \sim 8$ –10, $\log(\text{SFR}/M_\odot \text{ yr}^{-1}) \sim -2$ –1, and $\log(\text{sSFR}/\text{Gyr}^{-1}) \sim -1$ –0, respectively, are folded in the n_e range of ~ 30 – 100 cm^{-3} . Note that the sSFR values of the SFMS galaxies at $z \sim 0$ –1 are lower than those of the JWST galaxies. The SFMS galaxies at $z \sim 1$ –3 typically have $n_e \sim 100$ – 300 , while the parametric ranges examined are relatively narrow, i.e., $\log(M_*/M_\odot) \sim 10$ – 10.5 , $\log(\text{SFR}/M_\odot \text{ yr}^{-1}) \sim 1$ –2, and $\log(\text{sSFR}/\text{Gyr}^{-1}) \sim 0$ –0.5.

To cover higher sSFRs that match those of the JWST galaxies, we also refer to active star-forming galaxies at $z \sim 0$ (cyan diamond; Berg et al. 2022) and $z \sim 2$ –3 (yellow pentagon; Christensen et al. 2012a, 2012b; Sanders et al. 2016b; Gburek et al. 2019) whose sSFR values are comparable to those of the JWST galaxies (see Figure 3). We refer to these galaxies as high-sSFR galaxies. The median values of the high-sSFR galaxies at $z \sim 0$ and $z \sim 2$ –3 are ~ 100 and $\sim 200 \text{ cm}^{-3}$, respectively, both of which are comparable to those of the SFMS galaxies at a given redshift.

The left and center panels of Figure 4 illustrate that the n_e values increase from $z \sim 0$ –1 to $z \sim 1$ –3 and $z \sim 4$ –9 at a given M_* and SFR. This indicates that n_e evolves with redshift beyond the range that can be explained by the correlation between n_e and SFR. On the other hand, the sSFR distribution of the JWST galaxies does not overlap with that of the SFMS galaxies at $z \sim 0$ –3 as shown in the right panel of Figure 4, which makes it hard to tell if the n_e values evolve with redshift or correlate with sSFR. However, compared with the high-sSFR galaxies at $z \sim 0$ (cyan diamond) and $z \sim 2$ –3 (yellow pentagon), the n_e values are likely to increase from $z \sim 0$ –1 to $z \sim 1$ –3 and $z \sim 4$ –9 at a given sSFR as well. Note that the $z \sim 0$ high-sSFR sample consists of 45 galaxies, comparable to that of Davies et al. (2021) in each redshift bin, while the $z \sim 2$ –3 high-sSFR sample consists of only five galaxies due to the lack of statistical surveys for $z \sim 2$ –3 dwarf galaxies.

Figure 5 clearly shows that the n_e values typically increase from $z \sim 0$ to $z \sim 1$ –3 and $z \sim 4$ –9. We then investigate the

dependence of n_e on z by comparing the n_e values of the JWST galaxies with an extrapolation of the increasing trend identified in the lower- z galaxies. Assuming that the lower- z galaxies follow the $n_e \propto (1+z)$ or $n_e \propto (1+z)^2$ relation, we fit the two functions to the $z \sim 0$ –3 galaxies based on χ^2 minimization. We find that the JWST galaxies at $z \sim 4$ –9 have median n_e and z values (large red circles) falling in the gap between the $n_e \propto (1+z)$ and $n_e \propto (1+z)^2$ relations (gray shaded region), which suggests that galaxies at $z \sim 0$ –9 have an evolutionary relation of $n_e \propto (1+z)^p$ with $p \sim 1$ –2.

Although it is not obvious that global galaxy properties have an impact on the ISM property of n_e , the $n_e \propto (1+z)^2$ relation suggests that the nebula densities of high- z star-forming galaxies are generally high due to the compact morphology, i.e., smaller effective radii r_e and virial radii r_{vir} . Given the size evolutions of $r_e \propto (1+z)^{-1}$ (Shibuya et al. 2015; Ono et al. 2023) and $r_{\text{vir}} \propto (1+z)^{-1}$ (e.g., Mo & White 2002), stellar and halo mass densities are expected to be $r_e^{-2} \propto (1+z)^2$ and $r_{\text{vir}}^{-2} \propto (1+z)^2$ at a given mass, respectively, under the assumption that star-forming galaxies have disks (suggested by their low Sérsic indices; see Shibuya et al. 2015; Ono et al. 2023) whose heights are constant. If the stellar mass and halo mass densities are proportional to n_e , n_e is approximately proportional to $(1+z)^2$. This suggestion is similar to the conclusion claimed by Davies et al. (2021).

However, the JWST galaxies generally have n_e values slightly lower than those predicted by the $p = 2$ evolutionary relation. Although this discrepancy may be explained by the $\sim 20\%$ uncertainty of the electron temperature (Section 3.3), we should think of other factors that can decrease n_e values. One possible explanation is that thermal expansions at higher electron temperatures of high- z metal-poor nebulae result in lower n_e values for a given pressure (Kewley et al. 2019). The $p \sim 1$ –2 evolutionary relation of n_e may originate from a combination of the compact morphology and the high electron temperature of high- z galaxies. It should be noted that the JWST galaxies have M_* values ~ 1 dex smaller than those of the SFMS galaxies at $z \sim 0$ –3. Since the observed size–mass relation is a positive trend, $z \sim 0$ –3 galaxies with similar M_* to the JWST galaxies may have larger n_e values than the current

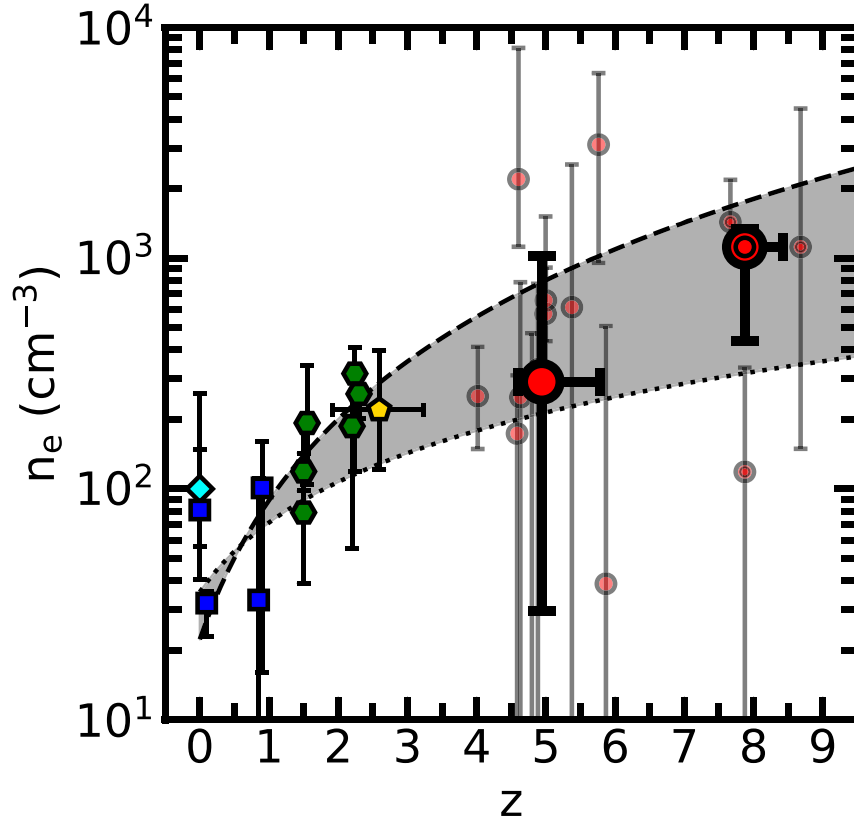


Figure 5. Electron density n_e as a function of redshift. All of the measurements in this figure are based on low-ionization lines (i.e., [O II] or [S II]). The symbols are the same as those in Figure 4. The dotted and dashed curves represent n_e following the $\propto(1+z)$ and $\propto(1+z)^2$ relations, respectively, which are determined by the galaxies at $z \sim 0$ –3. The JWST galaxies lie on the gap between the two functions (i.e., $n_e \propto (1+z)$ and $(1+z)^2$) filled with gray shading.

SFMS samples. This could lead to an even flatter redshift evolution of n_e .

It should also be noted that there is a possibility that the relatively high S/N criterion for the [O II] doublet (Section 2) causes a bias toward lower [O III] $\lambda 5007$ /[O II] $\lambda \lambda 3727, 3729$ ratios (O32, hereafter). This potential bias leads to ionization parameters q that are lower (e.g., Kewley & Dopita 2002), which lead to high n_e evidenced with $q \propto n_H^{-1}$, where n_H is a hydrogen number density roughly proportional to n_e . However, the JWST galaxies already have a high median O32 value of $\simeq 5$, which indicates that the JWST galaxies are not biased toward low O32.

5. Summary

We present the electron densities of singly ionized regions $n_e(\text{O II})$ in the ISM of SFMS galaxies at $z = 4$ –9 observed by the JWST/NIRSpec ERO, GLASS, and CEERS programs. Deriving LSFs of the NIRSpec instrument from in-flight calibration data, we securely measure the [O II] $\lambda 3726$ /[O II] $\lambda 3729$ ratios of 14 galaxies at $z = 4.02$ –8.68. We find that the 14 galaxies have $n_e \gtrsim 300 \text{ cm}^{-3}$ significantly higher than those of low- z galaxies at a given stellar mass, SFR, and sSFR (Figure 4). We also identify an increase in typical n_e values from $z = 0$ to $z = 1$ –3 and $z = 4$ –9, which is approximated by $n_e \propto (1+z)^p$ with $p \sim 1$ –2 (Figure 5). Although it is not obvious that the ISM property of n_e is influenced by global galaxy properties, the $p \sim 1$ –2 evolutionary relation can be explained by a combination of the compact morphology and the reduction of n_e due to the high electron temperatures of high- z metal-poor nebulae.

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ORCID iDs

Yuki Isobe  <https://orcid.org/0000-0001-7730-8634>
 Masami Ouchi  <https://orcid.org/0000-0002-1049-6658>
 Kimihiko Nakajima  <https://orcid.org/0000-0003-2965-5070>
 Yuichi Harikane  <https://orcid.org/0000-0002-6047-430X>
 Yoshiaki Ono  <https://orcid.org/0000-0001-9011-7605>
 Yi Xu  <https://orcid.org/0000-0002-5768-8235>
 Yechi Zhang  <https://orcid.org/0000-0003-3817-8739>
 Hiroya Umeda  <https://orcid.org/0009-0008-0167-5129>

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